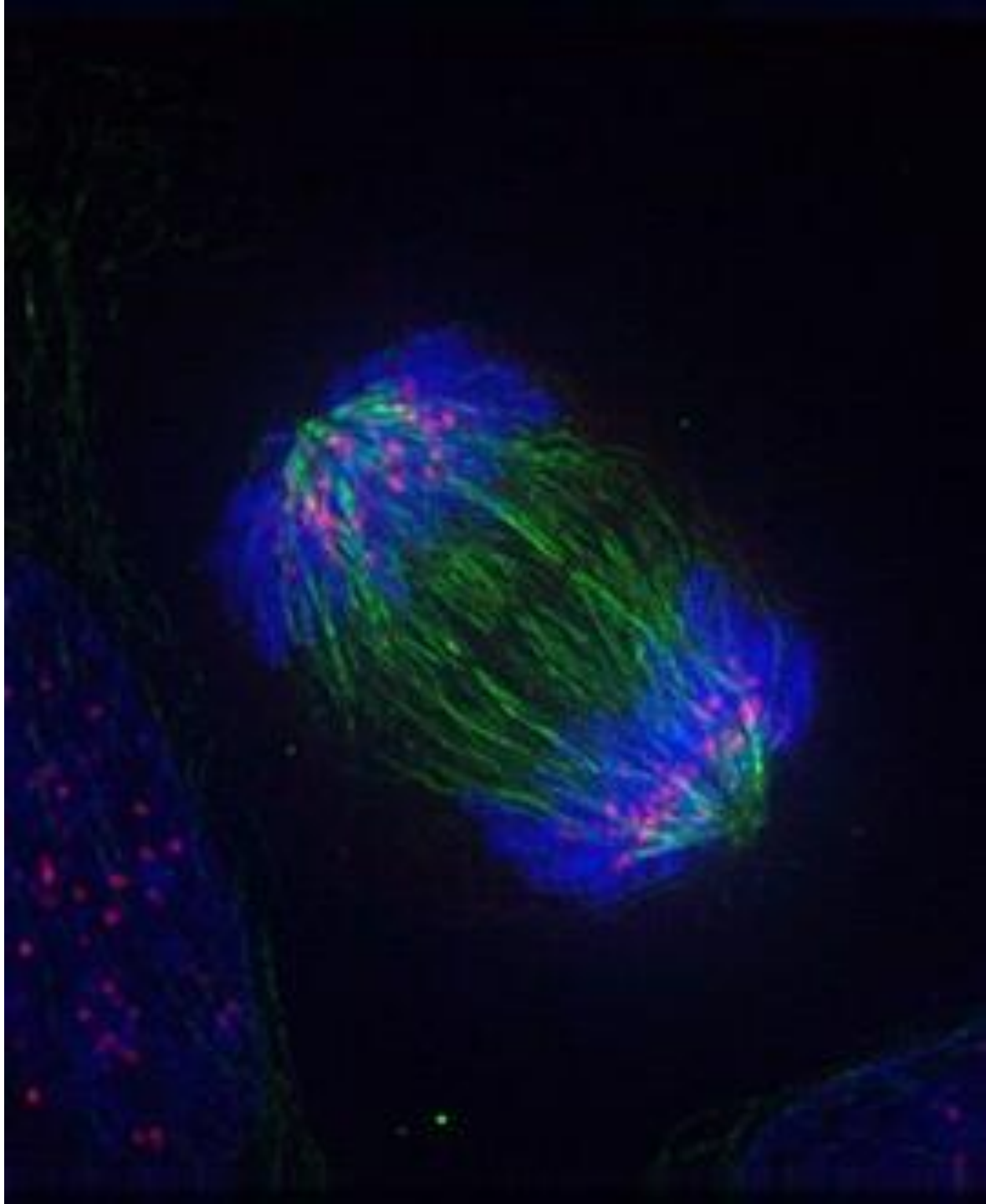
DP Biology – D2.1

Cell and nuclear division

- How are large numbers of cells with identical genes produced?
- How do eukaryotes make genetically varied cells that can become gametes?

SL statements D2.1.1–11. Textbook pp. 648–661.
Image: a cell in anaphase, stained. Roy van Heesbeen, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D2.1.1–5	Explain why cells divide, the roles of mitosis and meiosis, why DNA is replicated first, and how cytokinesis differs
2	D2.1.6–8	Describe how chromosomes condense and move, and identify the phases of mitosis
3	D2.1.9–11	Explain meiosis as a reduction division, non-disjunction, and how meiosis generates variation

New cells come from cell division

D2.1.1

A **parent cell** divides into two **daughter cells**. New cells only ever come from existing cells.

Growth

More cells in a multicellular body.

Repair and upkeep

New cells close wounds and replace worn-out cells.

Reproduction

New individuals, or gametes for sexual reproduction.

Key idea

In eukaryotes the nucleus divides first, then the cytoplasm (cytokinesis).

What mitosis and meiosis each do

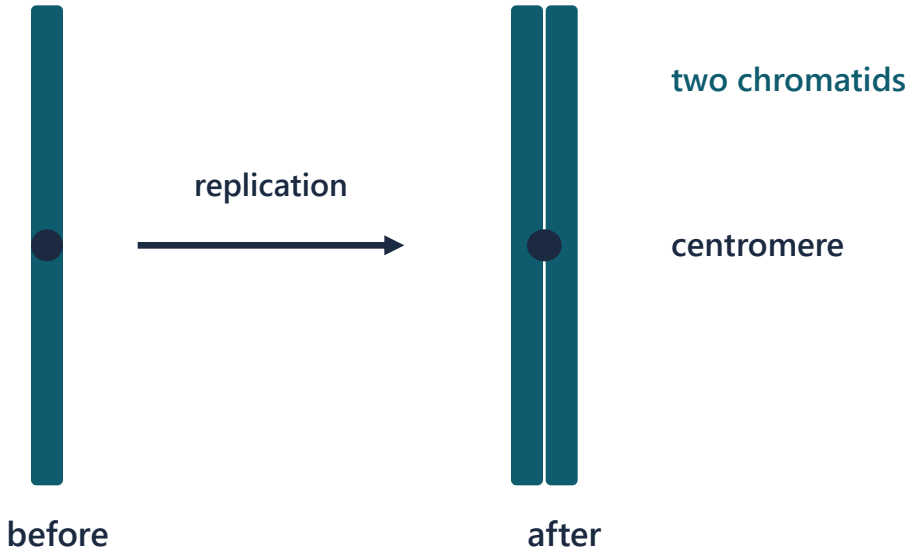
D2.1.4

	Mitosis	Meiosis
Chromosome number	Kept the same	Halved
Daughter nuclei	Identical	Genetically different
Divisions	One	Two
Used for	Growth, repair, asexual reproduction	Gametes in sexual life cycles

Why divide the nucleus first?
So each daughter cell gets a nucleus. A cell without one cannot keep making its proteins.

DNA is replicated before division

D2.1.5



DNA is copied in **interphase**. Each chromosome then has two identical DNA molecules, the **chromatids**, held at a **centromere**.

So each new nucleus can get a full set of genes.

Common mistake

After replication the chromosome number is unchanged; each chromosome has two chromatids.

Cytokinesis: splitting the cytoplasm

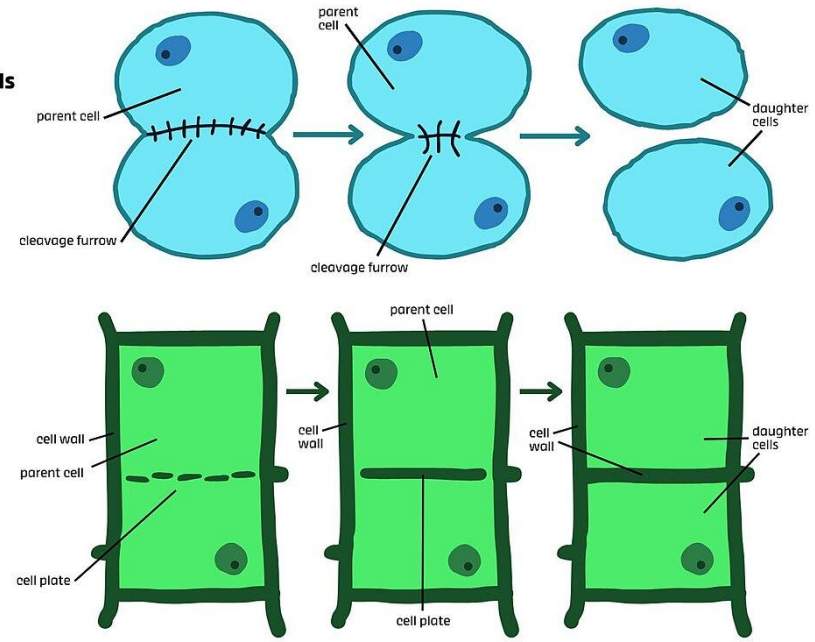
D2.1.2

Cytokinesis in animal cells

A cleavage furrow forms and is pinched inward until the parent cell splits into two daughter cells.

Cytokinesis in plant cells

A cell plate forms and expands to form a barrier between the two nuclei, forming into a cell wall and splitting the parent cell into two daughter cells.



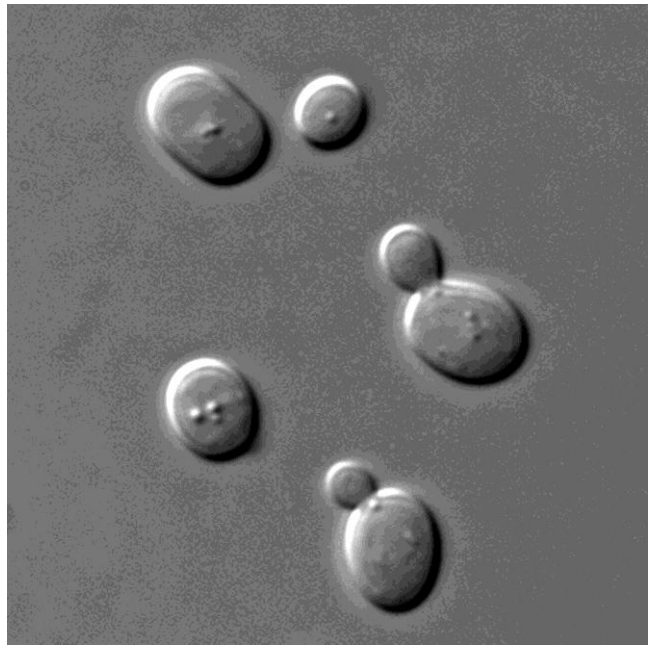
Cell	How it splits
Animal	A ring of **actin and myosin** pinches the membrane in
Plant	**Vesicles** at the equator build new membrane and wall

Cytokinesis in animal and plant cells. John Brian Eatmon, CC BY 4.0.

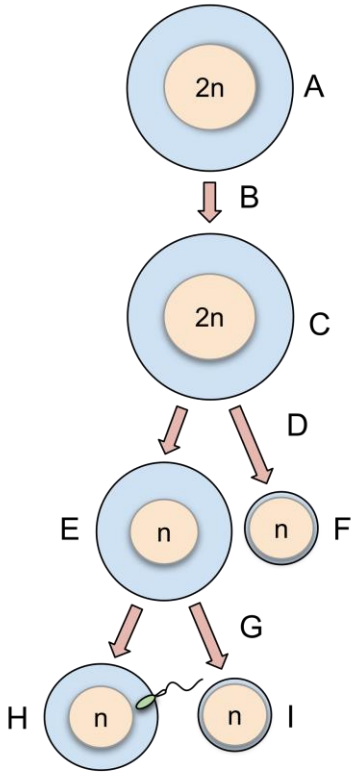
Why different?

Plant cells have rigid walls, so they cannot pinch in. They build a new wall from the centre outwards.

Equal and unequal cytokinesis



Budding yeast. Masur, public domain.



Oogenesis. Leiladavids, CC BY-SA 4.0.

Usually the cytoplasm is shared **equally**. In **oogenesis** the egg keeps most of it; in **yeast budding** a small bud forms.

Each daughter cell still needs at least one **mitochondrion**.

Think

Why is it useful for an egg to keep most of the cytoplasm?

Practice: count the chromosomes

D2.1.1–5

Stage of a human cell	Chromosomes	DNA molecules
Start of interphase	46	?
After replication	?	?
Each daughter after mitosis	?	?
Each nucleus after meiosis	?	?

1. Complete the table.
2. Explain why the DNA count doubles but the chromosome count does not.

Check-in: class 1

D2.1.1–5

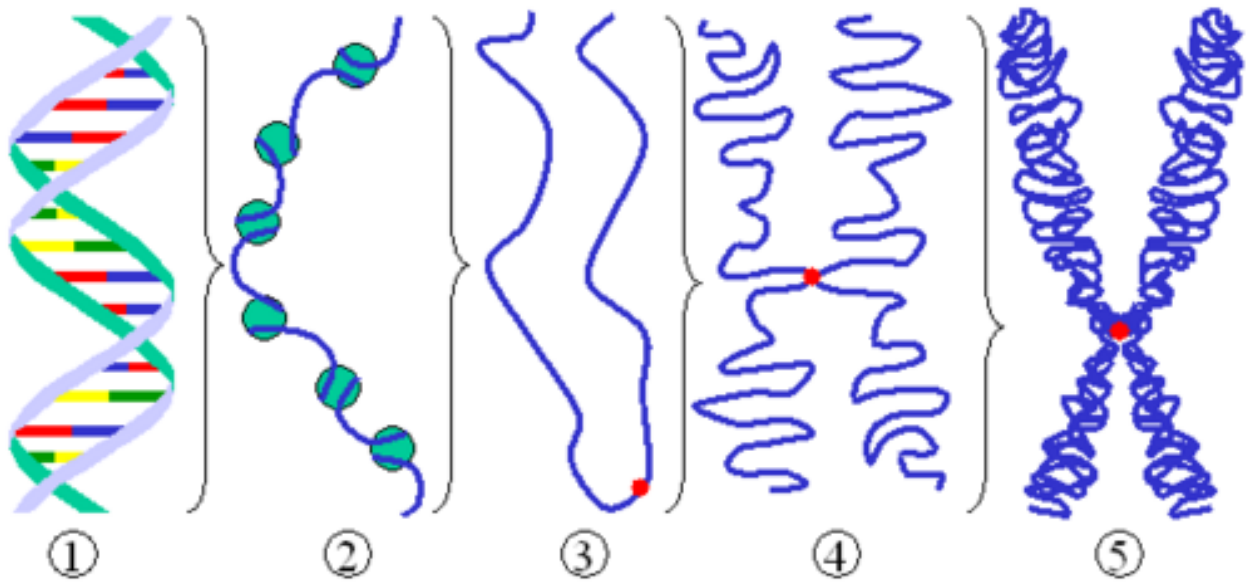
1. Give three reasons why cells divide.
2. Compare the outcomes of mitosis and meiosis.
3. Explain why DNA is replicated before division.
4. Distinguish between cytokinesis in animal and plant cells.
5. Give one example of unequal cytokinesis.

How to use this

Answer without your notes, then check against slides 3–8.

Condensation by supercoiling

D2.1.6



From a double helix to a chromosome. Magnus Manske, CC BY-SA 3.0.

In interphase DNA is long and thin. Before division it wraps around **histone** proteins and coils again and again: **supercoiling**.

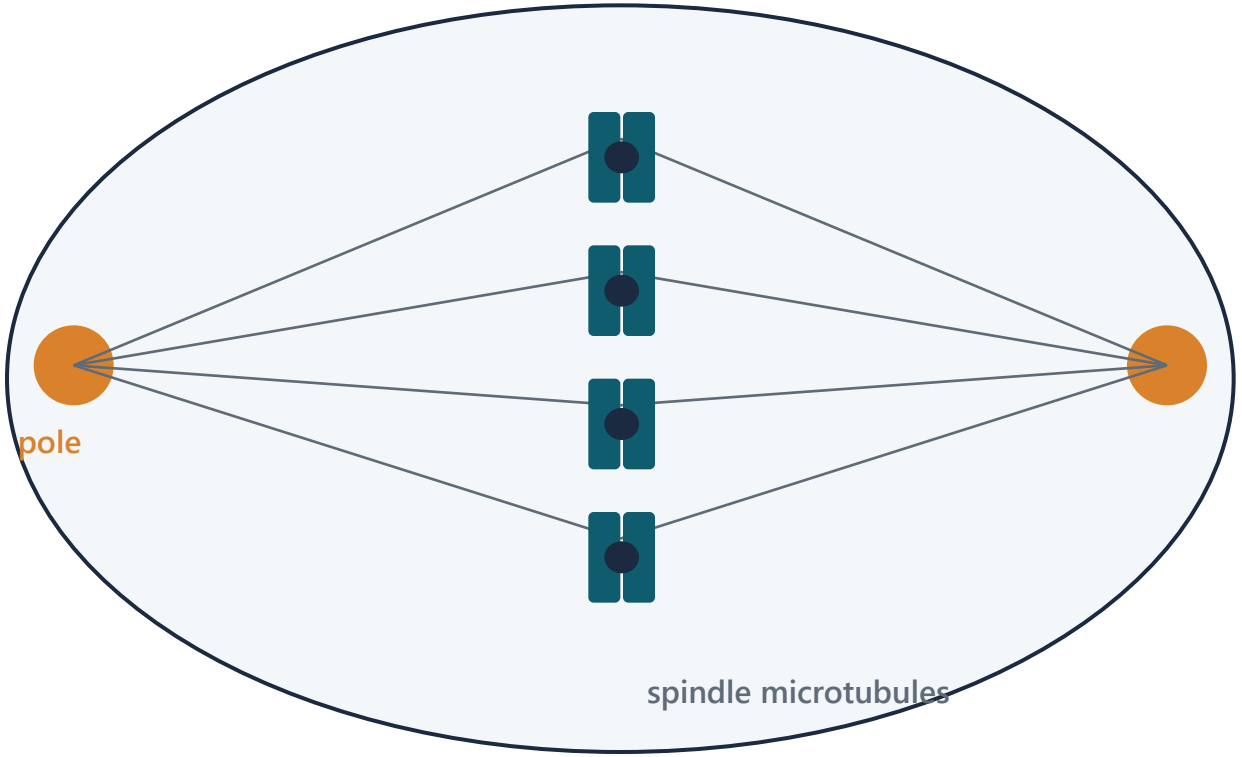
Short, compact chromosomes can be moved without tangling.

Scale

A human nucleus holds about 2 m of DNA. Condensed chromosomes are visible with a light microscope.

Moving chromosomes with microtubules

D2.1.6



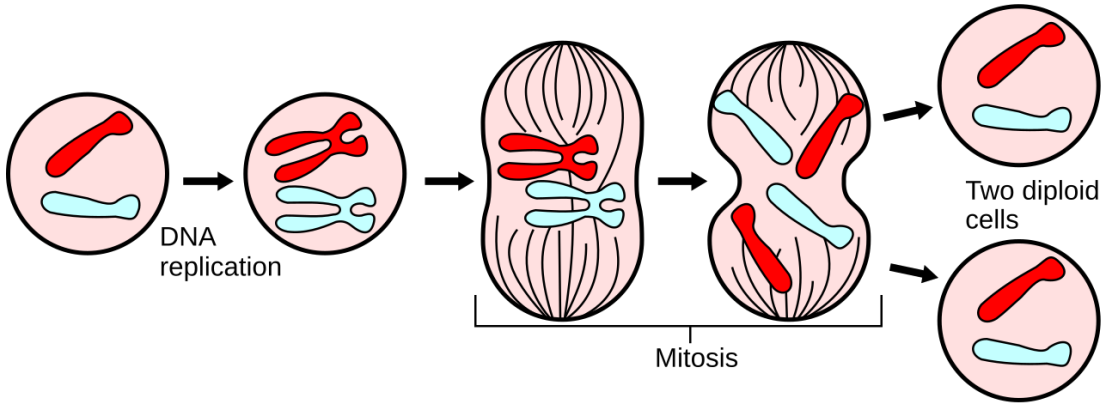
Term	Role
Microtubules	Protein tubes of the spindle
Centromere	Where the spindle attaches
Motor proteins	Use ATP to move chromosomes

Common mistake

Centromere, centrosome and centriole are different. So are chromatid and chromosome.

The phases of mitosis

D2.1.7

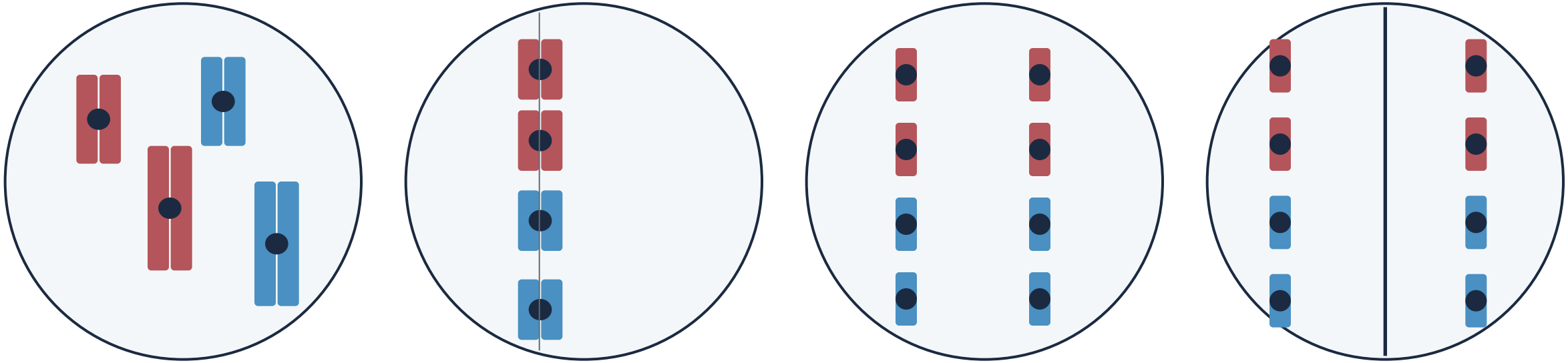


Major events in mitosis. Mysid, public domain.

Phase	What happens
Prophase	Chromosomes condense; nuclear membrane breaks down; spindle forms
Metaphase	Chromosomes line up at the equator; spindle attached to each centromere
Anaphase	Sister chromatids separate and move to opposite poles
Telophase	Nuclear membranes reform; chromosomes uncoil

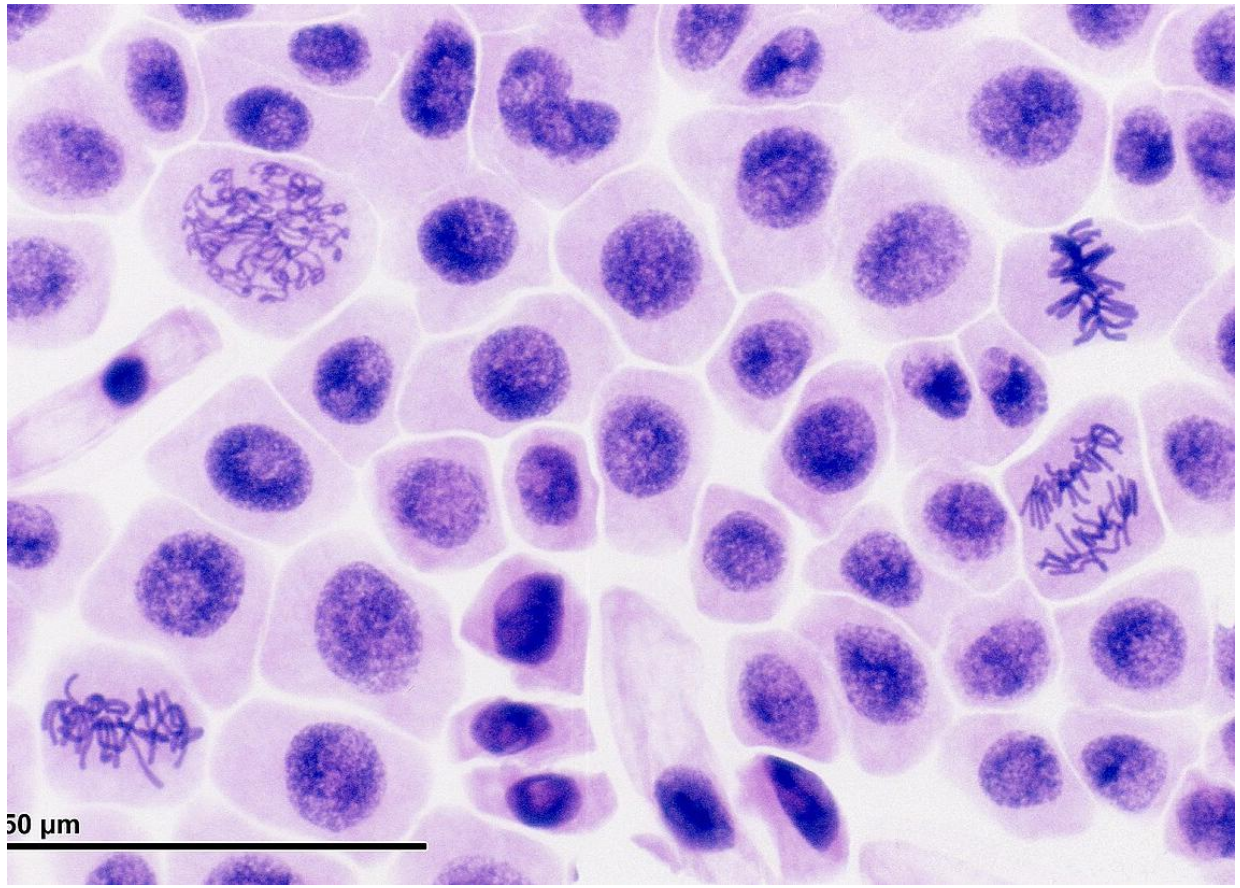
Mitosis in a cell with four chromosomes

D2.1.7



In anaphase each chromosome's two chromatids go to **opposite poles**, so each new nucleus gets one copy of every chromosome: the daughter nuclei are **identical**.

Identifying the phases of mitosis



Onion root tip cells in mitosis. Josef Reischig, CC BY-SA 3.0.

- **Prophase:** threads visible, no clear line
- **Metaphase:** a line at the centre
- **Anaphase:** two groups moving apart
- **Telophase:** two groups at the poles, new wall

Mitotic index

Cells in mitosis / total cells. If 18 of 150 cells are dividing, the index is 18 / 150 = 0.12.

Working with data: mitotic index

D2.1.8

Cells counted in two regions of an onion root. These counts are invented for practice; the pattern is realistic.

Region	Interphase	Prophase	Metaphase	Ana + telophase
Root tip	120	28	12	20
2 cm from tip	190	6	2	2

1. Calculate the mitotic index for each region.
2. Explain the difference.
3. Suggest why prophase has the most cells of any phase.

Check-in: class 2

D2.1.6–8

1. Explain why chromosomes condense before division.
2. State the role of microtubules.
3. Describe what happens in each phase of mitosis.
4. Explain why the daughter nuclei are identical.
5. Calculate the mitotic index if 24 of 200 cells are dividing.

How to use this

Answer without your notes, then check against slides 10–15.

Meiosis is a reduction division

D2.1.9

Term	Meaning
Diploid (2n)	Two sets of chromosomes, in homologous pairs. Human body cells: 46
Haploid (n)	One set. Human gametes: 23
Homologous	Same genes in the same order; alleles may differ

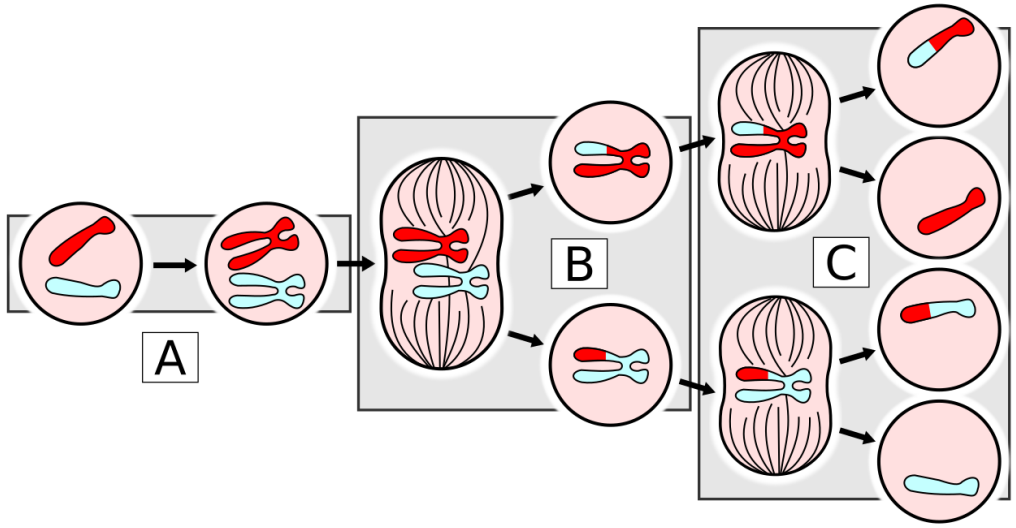
Meiosis turns one diploid nucleus into **four haploid nuclei**. Fertilization doubles the number; meiosis halves it again.

Common mistake

Each haploid nucleus gets one chromosome of every pair, not a random half of the DNA.

Two rounds of segregation

D2.1.9



Major events in meiosis. PatriciaR, public domain.

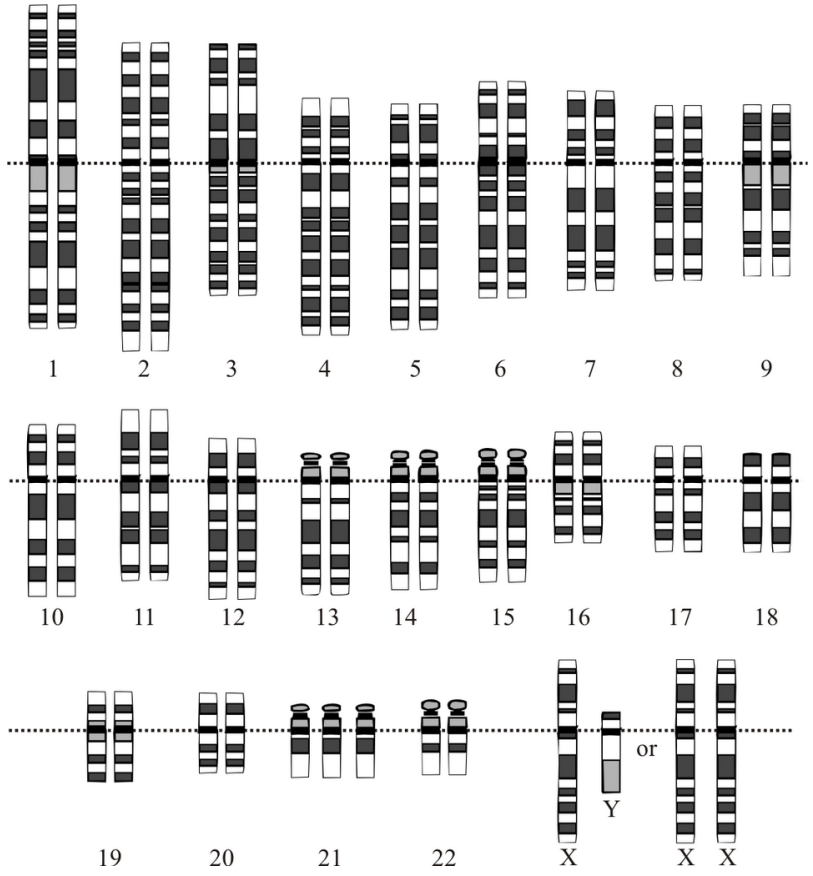
	Meiosis I	Meiosis II
Separates	**Homologous chromosomes**	**Sister chromatids**
Result	Two haploid nuclei; chromosomes still have two chromatids	Four haploid nuclei; one chromatid each

Comparing mitosis and meiosis

D2.1.9

	Mitosis	Meiosis
DNA replicated first?	Yes	Yes, once
Homologues pair up?	No	Yes, in prophase I
Nuclei made	Two	Four
Chromosome number	Same	Halved
Daughter nuclei	Identical	Different
Shared	Condensation, spindle	Condensation, spindle

Non-disjunction and Down syndrome



Karyotype with three copies of chromosome 21. NHGRI, public domain.

Non-disjunction: a pair of chromosomes (or chromatids) fails to separate, so one gamete gets an extra chromosome and another one fewer.

A zygote with three copies of chromosome 21 has **Down syndrome**.

Think

Does a higher risk at an older maternal age mean an older mother will have a child with Down syndrome?

Working with data: maternal age

D2.1.10

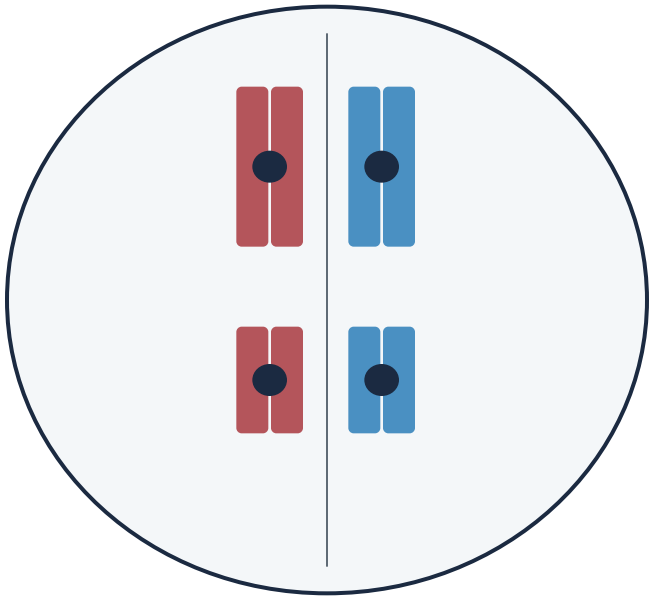
Approximate risk of trisomy 21 per live birth by age of mother (published estimates, rounded).

Age of mother	Risk
25	1 in 1250
30	1 in 1000
35	1 in 400
40	1 in 100
45	1 in 30

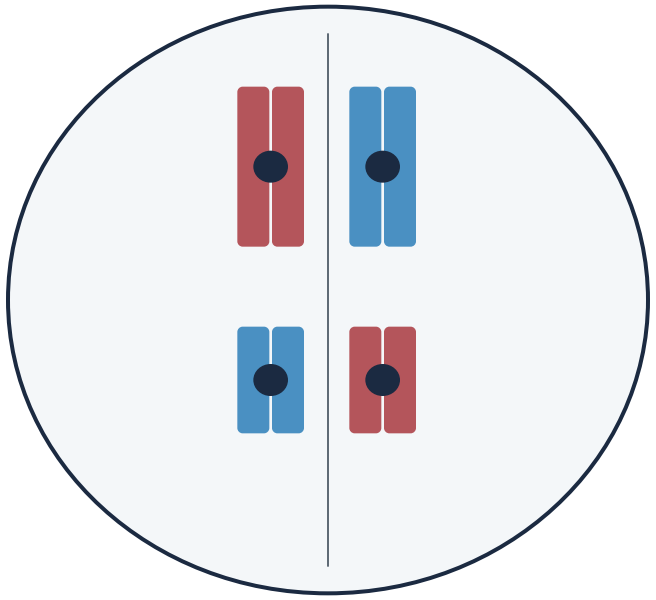
1. Describe the trend.
2. Calculate how many times greater the risk is at 40 than at 30.
3. Suggest why the risk rises with age.

Variation: random orientation

D2.1.11



one way



or the other

Drawn, n = 2. Red: from mother. Blue: from father.

In metaphase I each **bivalent** (pair) lines up at random, independently of the others.

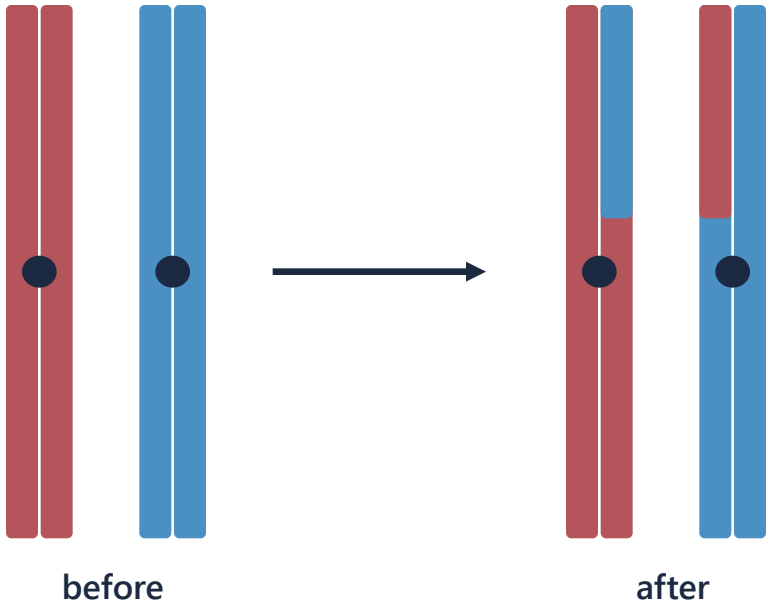
With **n** pairs: 2 to the power **n** combinations. Humans: over 8 million.

Key idea

Then random fertilization multiplies the combinations again.

Variation: crossing over

D2.1.11



In prophase I homologues pair up. **Non-sister chromatids** break and swap sections.

Chromatids now carry **new combinations of alleles**.

Common mistake

Crossing over makes new combinations, not new alleles. Swapping between identical sister chromatids would change nothing.

Practice: meiosis problems

D2.1.9–11

Count

2n = 8. How many chromosomes in each nucleus after meiosis?

Combine

n = 4. How many combinations from random orientation?

Error

A gamete has 24 chromosomes. Name the error and its cause.

Then

Explain why two children of the same parents (not identical twins) differ genetically.

Check-in: class 3

D2.1.9–11

1. Define diploid and haploid.
2. State what separates in meiosis I and in meiosis II.
3. Explain how non-disjunction leads to Down syndrome.
4. Explain how random orientation creates variation.
5. Explain how crossing over creates variation.

How to use this

Answer without your notes, then check against slides 17–24.

Common mistakes to avoid

Not this	But this
Chromosomes replicate in prophase.	DNA is replicated in interphase.
Homologues separate in mitosis.	Sister chromatids separate in mitosis; homologues in meiosis I.
Meiosis produces two cells.	Meiosis produces four haploid cells.
Cytokinesis is part of mitosis.	Mitosis divides the nucleus; cytokinesis the cytoplasm.
Down syndrome is a gene mutation.	It is an extra chromosome 21.

Summary: cell and nuclear division

	Key idea
D2.1.1–3	Division for growth, repair, reproduction; cytokinesis equal or unequal
D2.1.4–5	Mitosis keeps, meiosis halves; DNA copied first
D2.1.6	Supercoiling; spindle microtubules move chromosomes
D2.1.7–8	Prophase, metaphase, anaphase, telophase; mitotic index
D2.1.9	Meiosis I separates homologues; II separates chromatids
D2.1.10	Non-disjunction: trisomy 21
D2.1.11	Random orientation and crossing over

Check that you can...

1. **Compare** the roles of mitosis and meiosis
2. **Explain** why DNA replication comes before division
3. **Describe** the phases of mitosis and **identify** them in micrographs
4. **Distinguish** between equal and unequal cytokinesis
5. **Explain** meiosis as a reduction division
6. **Explain** non-disjunction and how meiosis generates variation

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. A cell has 2n = 8. State how many chromosomes each daughter nucleus has after mitosis, and after meiosis.

2. In which phase of mitosis do chromatids separate?

3. Give two ways in which meiosis generates genetic variation.

Image credits

Anaphase IF.jpg – Roy van Heesbeen – Public domain

Cytokinesis in animal cells and plant cells.jpg – John Brian Eatmon – CC BY 4.0

S cerevisiae under DIC microscopy.jpg – Masur – Public domain

Oogenesis.svg – Leiladavids – CC BY-SA 4.0

Chromatin chromosome.png – Magnus Manske – CC BY-SA 3.0

Major events in mitosis.svg – Mysid – Public domain

Mitosis (261 13) Pressed; root meristem of onion (cells in prophase, metaphase, anaphase, telophase).jpg – Doc. RNDr. Josef Reischig, CSc. – CC BY-SA 3.0

MajorEventsInMeiosis variant int.svg – internationalization: PatríciaR. Original image from NCBI; o – Public domain

Down Syndrome Karyotype.png – Courtesy: National Human Genome Research Institute – Public domain

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